

MARCOS JUAREZ, Argentina

WMO: 874670

Lat: 32.6797S

Lon: 62.1506W

Elev: 114

StdP: 99.96

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.2	0.4	-5.6	2.4	4.2	-4.1	2.7	5.1	15.0	11.0	12.8	11.1	1.2	50	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	34.4	24.0	33.1	23.3	31.8	22.8	26.2	31.7	25.2	30.6	24.3	29.3	4.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.0	29.8	23.7	18.8	28.4	22.8	17.7	27.3	82.7	31.9	78.2	30.7	74.2	29.5	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	10.5	9.0	-4.5	37.2	1.6	1.3	-5.6	38.2	-6.5	39.0	-7.4	39.7	-8.5	40.7		
(5)			-5.1	28.2	1.5	0.9	-6.2	28.8	-7.0	29.4	-7.9	29.9	-8.9	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	23.7	22.6	21.1	18.0	14.4	11.2	10.3	12.5	14.7	18.0	21.0	23.1
	DBStd	5.84	2.74	3.11	3.32	3.73	3.84	3.70	3.90	4.46	3.95	3.54	3.21	2.96
	HDD10.0	103	0	0	0	1	6	27	42	23	6	0	0	0
	HDD18.3	1041	1	3	12	51	134	217	251	191	120	48	12	2
	CDD10.0	2851	426	353	344	239	142	62	52	99	148	248	331	407
	CDD18.3	748	168	123	98	39	12	3	2	9	12	38	92	151
	CDH23.3	7305	1644	1078	872	337	77	11	18	113	174	446	986	1549
	CDH26.7	2837	693	423	317	92	13	1	4	39	46	147	388	674
Wind	WSAvg	3.4	2.9	2.7	2.8	3.1	2.9	3.2	3.5	4.0	4.3	4.3	3.9	3.6
Precipitation	PrecAvg	884	116	101	117	80	37	30	23	22	51	96	102	139
	PrecMax	1312	350	373	269	254	112	184	116	94	194	299	218	368
	PrecMin	597	9	28	15	1	0	0	0	0	1	12	18	38
	PrecStd	191	67	66	67	56	32	40	26	23	47	58	53	77
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.7	35.0	34.0	31.7	28.4	24.9	26.0	31.3	31.1	33.6	34.9	36.1
		MCWB	25.6	25.8	24.0	23.2	21.1	19.8	18.9	20.0	20.6	22.2	22.6	24.6
	2%	DB	33.7	32.9	31.9	29.3	25.4	21.1	21.9	26.2	27.8	30.5	32.8	34.1
		MCWB	24.7	24.6	23.1	21.4	19.4	16.6	16.1	18.1	18.4	20.3	21.2	23.5
	5%	DB	32.2	31.2	30.1	27.1	23.0	19.0	19.4	22.7	25.2	27.9	31.0	32.4
		MCWB	24.0	24.1	22.4	19.8	18.1	15.3	14.5	16.0	16.7	18.9	20.2	22.6
	10%	DB	30.6	29.4	28.3	25.0	20.8	17.3	17.3	20.2	22.9	25.6	28.9	30.6
		MCWB	23.3	23.0	21.3	19.1	16.9	13.9	12.8	14.1	15.5	17.8	19.6	21.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.5	27.5	26.4	24.7	22.4	20.2	19.6	21.3	21.9	24.0	24.6	27.2
		MCDB	33.0	33.1	32.0	29.7	26.2	23.6	24.4	28.7	29.0	31.2	31.9	33.2
	2%	WB	26.1	26.0	24.6	22.8	20.6	18.2	17.6	19.2	19.9	21.8	22.8	25.3
		MCDB	31.7	31.1	29.4	26.9	23.9	20.1	21.2	24.9	25.2	28.0	29.6	31.1
	5%	WB	25.1	24.8	23.3	21.4	19.1	16.6	15.7	17.2	18.2	20.3	21.6	24.1
		MCDB	30.4	29.5	28.3	25.1	22.0	18.1	18.0	21.3	23.2	25.5	27.9	29.8
	10%	WB	24.1	23.7	22.2	20.1	17.5	14.9	14.1	15.3	16.6	19.0	20.6	22.9
		MCDB	28.9	28.1	26.7	23.7	20.3	16.4	16.1	18.8	21.2	24.2	26.8	28.6
Mean Daily Temperature Range	5% DB	MDBR	12.0	11.3	12.0	11.4	10.7	11.0	12.0	13.3	13.6	12.6	13.0	12.6
		MCDBR	14.4	14.0	14.3	14.5	12.9	12.7	15.0	16.5	17.3	16.3	16.6	15.6
		MCWBR	6.9	6.7	6.6	7.2	7.1	8.0	9.0	9.0	8.9	7.5	7.0	6.8
	5% WB	MCDBR	12.5	11.8	12.0	11.5	10.6	8.9	11.6	14.4	14.2	13.5	14.0	12.9
		MCWBR	6.9	6.4	6.2	6.3	6.4	6.6	7.7	8.4	8.3	7.4	7.0	6.7
Clear-Sky Solar Irradiance	taub		0.408	0.390	0.376	0.373	0.350	0.333	0.342	0.393	0.452	0.416	0.396	0.397
	taud		2.320	2.386	2.426	2.404	2.420	2.464	2.407	2.231	2.063	2.234	2.321	2.340
	Ebn at Noon		931	927	905	848	815	808	815	811	813	892	936	945
	Edh at Noon		137	124	111	102	89	80	89	119	157	143	136	135
All-Sky Solar Radiation	RadAvg		7.21	6.21	5.26	3.81	2.80	2.47	2.71	3.64	4.80	5.73	6.89	7.33
	RadStd		0.41	0.48	0.43	0.47	0.27	0.32	0.26	0.34	0.42	0.46	0.42	0.47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	+0.50	+0.73	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.23	+0.49	+0.55	N/A	N/A	N/A	N/A

Nomenclature: See separate page